Fiscal Year:	FY 2021	**Task Last Updated:**	FY 10/13/2021
PI Name:	Taylor, Doris Ph.D.		
Project Title:	Vascular Health in Space: MicroRNAs in Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	None		
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Comments:	Dr. Taylor's email is dtaylor@texasheart.org		
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	05/01/2017	**End Date:**	12/31/2020
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 12/31/2020 per NSSC information and F. Hernandez/ARC (Ed., 1/28/2020) NOTE: Extended to 12/31/2019 per NSSC information and F. Hernandez/ARC (Ed., 12/14/18) NOTE: End date changed to 10/31/2018 (original end date was 10/31/2017) per F. Hernandez/ARC (Ed., 10/20/17)		
Key Personnel Changes/Previous PI:	Ed. note, 10/13/2021: Project ended December 2020 and continues with Principal Investigator Sonja Schrepfer, MD PhD, University of California San Francisco, "Vascular Health in Space: MicroRNAs in Microgravity (PI: Schrepfer)", grant 80NSSC21K1521.		
COI Name (Institution):	Schrepfer, Sonja M.D., Ph.D. (University Medical Center Hamburg, Germany; University of California San Francisco (UCSF))		
Grant/Contract No.:	NNX15AB55G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continues "Vascular Health in Space: MicroRNAs in Microgravity" with Principal Investigator (PI) Robert Robbins, M.D., due to Dr. Robbins's move to University of Arizona. Dr. Doris Taylor is new PI as of May 1, 2017. Astronauts are faced with several health risks during both short- and long-duration spaceflight due to the space environment. Microgravity itself seems to affect cardiovascular parameters. For example, studies on animals (rats and rhesus monkeys) have found consistent decreases in heart rate during spaceflight. Similarly, data from 12 astronauts over 6 missions showed that heart rates of astronauts were reduced in space. These early studies demonstrate an obvious change in an easy to measure cardiovascular parameter. Although cardiovascular disease is the leading cause of death in the United States, to our knowledge, cardiovascular physiology in space has not been studied, yet. More in-depth studies are now required to shed light on how microgravity exerts changes on individual cells and tissues. We believe that subcellular effects of microgravity can be documented and we hypothesize that microRNAs (miRNAs), because of their regulatory function, will be involved. miRNAs are now understood to be powerful modulators of biological pathways and have generated considerable excitement as next-generation therapeutic targets for a variety of diseases, including cardiovascular diseases. In this proposal, we will assess the gravitational impact on the miRNA response to vascular injury and will evaluate the efficiency of vascular repair in space. This will shed light on how microgravity alters vascular remodeling and healing and could reveal currently unknown targets for potential therapeutic modulation. We will evaluate the role of miRNAs in vascular disease (Objective 1), vascular health (Objective 2), vascular smooth muscle cell function (Objective 3), and angiogenesis (Objective 4). Previously, we investigated miRNA regulation in diseased human vessels with myointimal lesions and demonstrated the important regulatory role of miRNAs in vessel injury. Our studies demonstrated that various vascular miRNAs were downregulated after vessel injury and only a few were upregulated. Recent spaceflight studies showed that the expression of miRNAs in both men and mice were sensitive to gravity. The limited data available demonstrated down-regulation of multiple miRNAs in microgravity. We therefore hypothesize that microgravity could aggravate pathologic vessel remodeling via miRNA alterations. Because inflammation is a known stimulus for myointimal hyperplasia, we will pay special attention to the inflammatory responses to vessel injury. In Objective 1, we will investigate how vascular miRNAs are differentially regulated at 1xg and Âµg in response to vascular balloon injury. In Objective 2, we will investigate the effect of microgravity on miRNAs in healthy vessels. If the gravitational effect on vascular miRNAs is profound, microgravity might even induce the development of vascular lesions in healthy vessels. In Objective 3, we will assess the effect of microgravity on smooth muscle cell (SMC) function in both healthy and diseased vessels. Under normal physiologic conditions, SMCs rarely proliferate. However, in the presence of injury or a growth stimulus, SMCs begin to proliferate and migrate. Microgravity could be another so far unknown growth stimulus. Uncontrolled SMC proliferation can result in pathologic vasoconstriction or in myointimal hyperplasia with narrowing of the lumen. Microgravity might thus affect SMC plasticity. In Objective 4, we will assess and compare angiogenesis at 1xg and µg and define the role of angiogenetic miRNAs. Since angiogenesis plays an important role in vascular health, we anticipate that effects of microgravity on angiogenesis-associated miRNAs will impact vascular health and repair processes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The expected gain in knowledge will also advance the research on vascular miRNAs on Earth. Our group and a few others have focused on altering miRNA expression in injured vessels using systemic or local delivery of anti-miRNAs. We believe that therapeutically altering miRNAs is a newly evolving strategy with enormous potential. The results we expect to achieve from this study will thus complement the work that has been done under static gravitation on Earth. They could provide insights into genetic regulation and signaling pathways and reveal new targets for therapeutic modulation, including potential new pharmaceutical products.
Task Progress:	Progress as of October 2020, reported to NASA Ames Research Center (received for Task Book September 2021): Ground/Preflight task summary: To investigate the effect of microgravity on vascular remodeling-associated miRNAs. We determined the effects of microgravity on miRNAs that are associated with vascular remodeling after injury using RNA sequencing and gene chip analysis (microarrays). We investigated the effect of microgravity on vascular remodeling-associated genes. We determined the effects of microgravity on genes that are associated with vascular remodeling after injury using RNA sequencing and gene chip analysis (microarrays). We investigated the effect of microgravity on vascular remodeling-associated proteins. We assessed the effect of microgravity on SMC plasticity using immunoblotting. We investigated the effect of microgravity on SMC morphology. We assess the effect of microgravity on SMC morphology using histopathology, electron microscopy, and 3D-confocal immunofluorescence. We investigated the effect of microgravity on SMC behavior. We assessed the effect of microgravity on SMC proliferation and apoptosis. Ed. note, 10/13/2021: Project continues with Principal Investigator Sonja Schrepfer, MD PhD, University of California San Francisco, "Vascular Health in Space: MicroRNAs in Microgravity (PI: Schrepfer)", grant 80NSSC21K1521. See that project for future reporting.
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